



HCI & CG Assignment

Name: Shamoo

Roll no: 2K24/CSE/141

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Task1:

Output: i.e Standard Monitor

```
--- DISPLAY METRICS ANALYSIS ---  
Total Pixel Count:  2,073,600 pixels  
Aspect Ratio:      16:9  
Calculated DPI:    91.79 DPI  
Density Category:  Low Density (Standard Monitor)
```

Output: i.e Smartphone Screen

```
Enter horizontal resolution (pixels): 1920  
Enter vertical resolution (pixels): 1080  
Enter physical diagonal size (inches): 6.1
```

```
--- DISPLAY METRICS ANALYSIS ---  
Total Pixel Count:  2,073,600 pixels  
Aspect Ratio:      16:9  
Calculated DPI:    361.13 DPI  
Density Category:  High Density (Retina / Mobile)
```

Task2:

Image Created:



Output:

```
--- SYNTHETIC MATRIX METRICS ---  
Array Shape (H, W, C): (300, 400, 3)  
Data Type:      uint8  
Total Elements: 360,000 values  
Memory Footprint: 360,000 bytes (351.56 KB)
```

Task3:

Image:

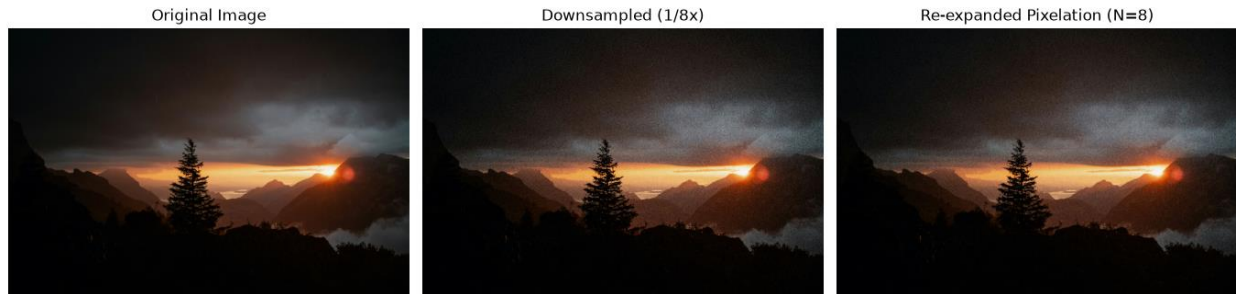


Output:

```
--- CHANNEL EXTRACTION SUMMARY ---  
Original Image Shape: (4160, 6240, 3)  
Red Channel Shape:    (4160, 6240) | Mean Intensity: 42.12  
Green Channel Shape:  (4160, 6240) | Mean Intensity: 33.03  
Blue Channel Shape:   (4160, 6240) | Mean Intensity: 27.75
```

Task4:

Image:



Output:

```
--- DOWNSAMPLING ANALYSIS (N=8) ---  
Original Shape: (4160, 6240, 3) | Memory: 77,875,200 bytes  
Downsampled Shape: (520, 780, 3) | Memory: 1,216,800 bytes  
Re-expanded Shape: (4160, 6240, 3) | Visual: Blocky Pixelation  
Dimension Reduction: 87.50% reduction per axis  
Memory Savings: 98.44% data reduction
```

Question:

Why does adding an Alpha channel (RGBA) increase an image array's memory consumption by 33% compared to standard RGB?

RGB uses 3 channels, while RGBA adds a fourth channel for transparency. Adding one extra channel to three existing channels increases the data by one-third, which equals approximately 33 percent